

'Simply Irresistible': Recurring Accent Patterns as Hooks in Mainstream 1980s Music

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‘Simply Irresistible’: recurring accent patterns as hooks in mainstream 1980s music

DON TRAUT

Abstract

This essay explores connections between the notion of ‘hook’ and several recurring accent patterns found in rock music from the 1980s. It begins by identifying seven syncopated accent patterns and presents a list of over 150 songs from the 1980s that use one or more of these patterns. It then presents three main ways that the accent patterns help create ‘hooks’. First, several songs use one of the patterns to create a strong title hook, as in Rick Springfield’s hit, ‘I’ve Done Everything for You’, the title of which is set with a syncopated and distinct rhythmic pattern. A second type of ‘hook’ comes in the form of a ‘significant gesture’ that usually precedes the chorus. The pre-chorus ‘My-my-my-my, ah, woo!’ gesture from The Knack’s ‘My Sharona’ is an excellent example. Finally, a third type of ‘hook’ involves the combination of a repeated accent pattern with a recurring pitch pattern, as in the guitar introduction to AC/DC’s ‘For Those About to Rock’. By realising that these three examples all use the same accent pattern in significant, but different, ways, we acknowledge the prominence of this and other patterns while furthering the discourse on the notion of the ‘hook’.

Introduction

The notion of the ‘hook’ remains a troubling concept for scholars of popular music. What exactly does it mean to describe a gesture as a ‘hook’? Part of the problem with this term is that it can mean different things to different people. The term itself comes from the music industry and is associated with ‘catchiness’ and marketability. For the listener, the meaning is much the same; the hook is the most memorable line or part of a song, the part that first comes to mind when one thinks of that song. Of course, this can be problematic as well, since there is no guarantee that two different listeners will get ‘hooked’ by the same gesture. While the subjective nature of these notions of hook is not a problem for fans or those in the music industry, it does create headaches for music scholars. And when we go further and ask not only ‘What is a hook?’, but ‘What is it about the music that makes this a hook?’, the picture gets even more blurry.

Gary Burns addresses some of these questions in an excellent study titled, ‘A typology of “hooks” in popular records’ (Burns 1987, pp. 1–20). After considering several definitions and criteria, Burns proposes that hooks should be easily remembered musical events that stand out from what surrounds them (*ibid.*, p. 1). In the course of adopting this definition, he often downplays the role of repetition. First, he states that ‘repetition is not essential in a hook, but it is not ruled out either’, and later, he asserts that ‘repetition is meaningless without its opposite, change’ (*ibid.*, p. 1). Since he clearly considers change a key element in defining a hook, he adopts the term

'modulation' in a general way 'to refer to a major change in any structural element' (*ibid.*, p. 2). With this general definition in mind, Burns then presents numerous ways artists create hooks via change in various musical parameters.

Included in Burns' typology are 'rhythm hooks', or hooks that occur anytime there is a modulation in the rhythm. He begins with the most obvious change 'from no rhythm to some rhythm' right at the beginning of the song (*ibid.*, p. 6). Other rhythmic hooks are more internal, occurring within the song, be they drastic changes in the metre, or the simple addition or deletion of a beat. Whatever the case, he is careful to avoid the notion that repeated rhythmic figures could be recognised as hooks. For Burns, a musical gesture or element must stand out from its context to qualify as a hook. As he puts it then, 'rhythmic hooks are thus a result of change rather than repetition' (*ibid.*, p. 6).

Burns' notion of the rhythmic hook seems to fit nicely with Charles Keil's distinction between 'vital drive' and 'gesture' in jazz music. To quote Keil:

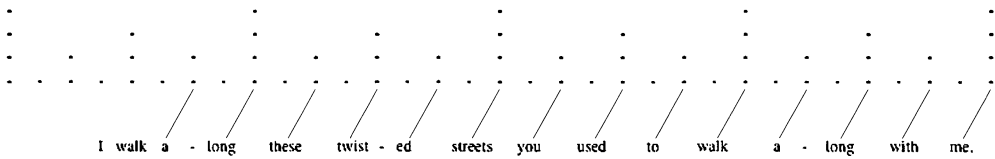
In jazz, it seems to me, the net effect of an entire piece may focus on one or two significant gestures; indeed, a vital drive may be seen as a device for holding our attention and increasing our involvement so that a single phrase, 'weighted just right', will have maximum impact. (Keil 1994, p. 72)

In general, Keil's points apply to rock music as well.¹ In fact, while their terminology is different, Keil and Burns appear to be addressing the same types of phenomena. What Keil calls a 'significant gesture' might well meet the criteria of a hook for Burns. Judging by Keil's comments, this gesture would typically be a specific moment or passage that has an impact when compared to the vital drive, which has been holding the listener's attention via repetition. This is in line with Burns' tenet that a hook must present some type of modulation away from the established context.

Despite these apparent affinities between Burns and Keil, there is still room for other interpretations of what constitutes a hook, especially where the role of repetition is concerned.² This paper argues that the apparent distinction between vital drive and hook may not be so clear, that the distinction is not always such an easy one to make. It does this first by identifying and studying the structure of several accent patterns that can be found in multitudes of songs. This discussion will highlight the important role that syncopation plays in this style, particularly as it pertains to creating tension against the normative pulse that underlies most rock music.

From there, the discussion turns to the distinction between 'vital drive' and 'hook', and the role of repetition. In this portion, the patterns are cast in a more contextual light, revealing how specific songs utilise the same pattern, but in different ways. In one song, a particular pattern may serve as a clear contrast to the rest of the song, creating a distinct hook. But in other cases this distinction is not so clear. Rhythmic patterns, despite being repeated throughout entire songs in some cases, can form strong hooks when they contain rhythmically interesting syncopation patterns, or when they are combined with recurring pitch patterns. For these reasons, it becomes very difficult to deny the 'catchiness' of these syncopated accent patterns, despite their repetition.

Since the patterns addressed herein may not be equally prevalent in all types of rock music, it seems wise to place some type of frame on the song sample. While the patterns discussed below do occur in a range of styles and eras, they are most prevalent in mainstream pop and rock songs from the 1980s. In many ways, this is not surprising. Entire radio stations and dance clubs base their play lists on this era,



Example 1. Syncopation in Naked Eyes' 'Always Something There to Remind Me'.

partially due to what Burns calls 'the rather homogeneous rhythm of MTV-style rock in the 1980s (epitomised by eight-to-the-bar, with a loud snare drum on three and seven)' (Burns 1987, p. 6). With this in mind, the bulk of this paper focuses on songs released during the 1980s. The paper's conclusion will suggest further pervasiveness of these patterns in other eras of rock music.

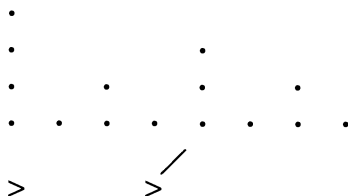
Structural aspects of the accent patterns

It is well known that syncopation plays an important role in most rock music. It is the main means of providing interest against the 'strong-weak' alternation of the underlying pulse. The relationship between this underlying pulse and the syncopation of the surface rhythm has received attention from other writers. Richard Middleton, for example, suggests that we explore 'transformations' in rock rhythm, 'so a simple writing of 4/4 can be transformed into its syncopated equivalent' (Middleton 1990, pp. 212–13). In this manner, Middleton argues, we can see how syncopation stems from the underlying pulse and is in fact, 'heard "with reference to", "in light of", "as a remapping of", its partner' (*ibid.*, p. 213).

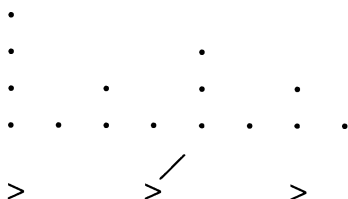
Since syncopation is such an important property of the rhythmic patterns discussed below, it will be especially important to keep clear this distinction between rhythm and underlying pulse. For that reason, this paper incorporates the representational system that Lerdahl and Jackendoff used in their book on tonal music (Lerdahl and Jackendoff 1983, ch. 2). In their system, levels of dots represent the underlying pulse of a 'well-formed' metric passage. Surface rhythms can then be superimposed on this grid and analysed for their adherence to or deviation from the grid.³ Example 1 shows how this system works. It also points to the prominence of syncopation in rock music. Each row of dots represents a level of structure (i.e. eighth-notes, quarter-notes, half-notes, whole-notes) in the underlying duple pattern of four measures in common time. Below this grid of dots is the opening verse of Naked Eyes' 1983 hit, 'Always Something There To Remind Me'. Superimposing the hierarchic grid on this passage highlights the specific rhythmic placement of each attack. Note that the text setting is almost entirely syncopated (as shown by the diagonal lines), continuously stressing weak, or 'off-beats', thus going against the stresses inherent in the 'well-formed' underlying structure.

Example 1 gives just one sample of the prominent role syncopation plays in rock music. It is part of the style, like *bel canto* singing in most opera, or the swing eighth-note in most jazz. That said, it would be difficult to argue that the syncopated nature of the melodic line was part of the hook of the above song. Certainly, the tune would be different without the syncopation and, in fact, would not really 'work', but the presence of syncopation alone does not constitute a hook. What Example 1 does *not* capture is the manner in which syncopation often forms short patterns that rock musicians employ in numerous ways. These shorter patterns use syncopation in a more distinct manner than the type shown in Example 1. Moreover, identical accent

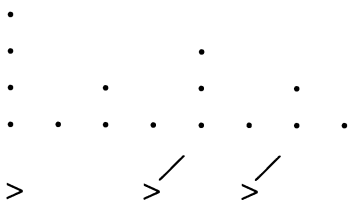
(a) < 3 5 >



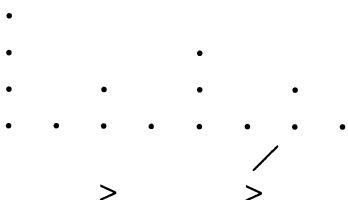
(b) < 3 3 2 >



(c) < 3 2 3 >



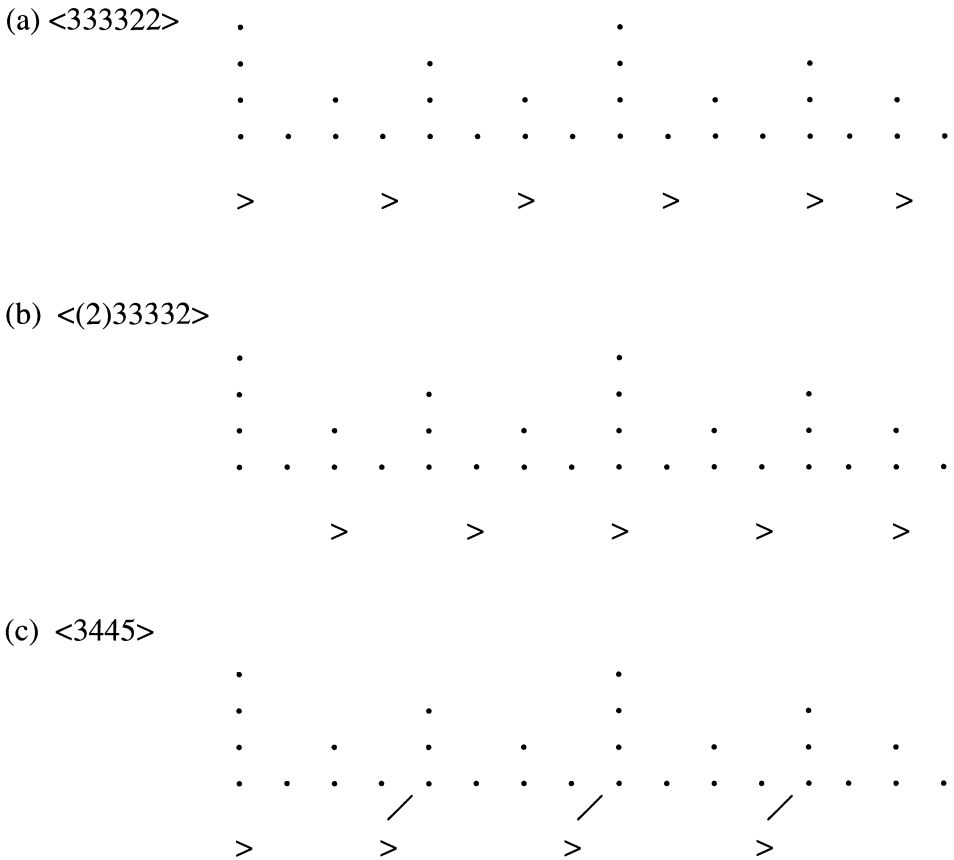
(d) < (2) 3 3 >



Example 2. Four common one-measure accent patterns.

patterns can be found in multitudes of songs, further suggesting the 'catchiness' of their appeal.

Example 2 presents four of these recurring accent patterns.⁴ Each pattern is one measure in length, meaning they group the eight eighth-notes of a single common-time measure into two or three groups. The number of eighth-notes between accents determines the size of these groups, represented by the Arabic numbers contained inside the brackets ('<' and '>'). Thus, in Example 2(a), a single syncopated accent on the 'and' of two divides the eight eighth-notes into two groups, one containing three eighth-notes, the other containing five. Example 2(a) is thus considered a <35>



Example 3. Three common two-measure accent patterns.

pattern. Each pattern contains at least one syncopation, represented by the diagonal line(s), which shows the displacement of the appropriate underlying beat. Examples 2(a) and (b) are quite similar, as each anticipates beat three.⁵ This accent brings a nice asymmetry to the measure, providing more motion to the upcoming downbeat. The only difference here is the accent of beat four in Example 2(b), which creates a <332> pattern that seemingly 'resolves' the metric dissonance created by the consecutive groups of three, and prepares the pattern to repeat.⁶

The remaining one-measure patterns are less common, but they do each occur in several songs. Examples 2(c)–(d) are closely related to Example 2(b) in that each contains two groups of three and one group of two. Interestingly, the arrangement of these groups completes all possible arrangements of the internal groups and, in fact, forms a complete rotation of the original <332> layout.⁷ Examples 2(c)–(d) *do* bring some unique qualities as well. Example 2(c), for instance, is the only pattern shown here with two syncopations, anticipating beats three and four; and in Example 2(d), no accent is shown on the downbeat. This is because in practice songs using this pattern leave the downbeat relatively unaccented, placing more emphasis on beat two.

Perhaps even more common than the patterns shown in Example 2 are those which encompass two measures of music, as shown in Example 3. In these patterns,

accents divide sixteen eighth-notes into groups of various sizes. As with the one-measure patterns, the accents here create rhythmic tension without obliterating the underlying duple feel. Example 3(a) shows the most common two-measure pattern. Here, four groups of three create a hemiola that lasts one-and-a-half measures before being rectified by the pair of two-groups, which signify the pattern's end and prepare for the arrival of the next measure. It results in only two true syncopations, but their impact is great.

The metric conflict created in this particular pattern touches on several points made by other authors. In his presentation on metric dissonance in rock music, for example, Scott Murphy featured it as a prime example of conflict and resolution, felt as the groups of three create tension before giving way to the two-groups, which seem to realign the rhythm with the underlying metre and 'resolve' the metric tension (Murphy 2002). And in his study on syncopation, David Temperley addresses the tension formed by extended syncopations such as this one. He draws a brief contrast between syncopation in rock and classical music, stating that while classical music often blurs the underlying pulse, 'metrical ambiguity is not common in most kinds of rock' (Temperley 1999, p. 35). He does acknowledge, however, that there are those passages that highlight the conflict between surface rhythm and underlying metre. As he states:

[T]he main point here is that rock encourages us to try to keep a metrical structure going even when there are stressed events that (on the surface) conflict with it; in genuinely irregular passages . . . , this can create metrical situations of considerable complexity and richness. (Temperley 1999, p. 37)

Temperley's words here are quite apt for the <333322> pattern because it does very much go against the underlying duple structure, but it also helps listeners get 'back on track' with the two-groups at the end. It is the tension of hearing the surface rhythm, yet sensing the underlying metre, that makes this pattern so effective.

Not surprisingly, as with the one-measure patterns, several songs feature variants of the two-measure <333322> pattern, variants that shift the accents and create somewhat different syncopations. The most common variant retains the four groups of three and two groups of two seen in the prototype, but it shows one very important difference: it rotates the pattern such that the four groups of three fall *between* the two groups of two, thereby creating a kind of hybrid <(2)33> and <332> pattern, as shown in Example 3(b). While there is a clear duple group that begins on the downbeat of the measure, in practice this initial two group is almost always unaccented (just as it was unaccented in the <(2)33> one-measure pattern seen earlier), placing heavy stress on beat two, where the first three-group begins. Thus, the true accents fall on beat 2, the 'and' of three, beat one of the second measure, and the 'and' of two. The sense of the rhythmic 'dissonance' being 'resolved' is not so strong here as it was in the <333322> pattern, but it is still in effect. Finally, Example 3(c) is not a direct variant of the <333322> prototype, but it is a quite prominent two-measure structure. This is an attractive pattern because it provides four accents over two measures, and all except the initial downbeat are syncopated. Beat three is anticipated in both measures, as is the downbeat of the second measure.

Accent patterns and the notion of 'hook'

Table 1 lists over 150 hits songs from the 1980s that contain one or more of the syncopated accent patterns presented in Examples 2 and 3. The Table gives the song

Table 1. 1980s songs featuring patterns from Examples 2 and 3.

Song	Year	Artist	Pattern
'Babylon Sisters'	'80	Steely Dan	<35>
'The Waiting'	'81	Tom Petty	<35>
'Karma Chameleon'	'83	Culture Club	<35>
'Billie Jean'	'83	Michael Jackson	<35>
'Burning Heart'	'85	Survivor	<35>
'Like a Virgin'	'84	Madonna	<35>
'Kiss On My List'	'81	Hall and Oates	<35>
'Carribean Queen'	'84	Billy Ocean	<35>
'Like a Rock'	'85	Bob Seeger	<35>
'I'll Wait'	'83	Van Halen	<35>
'Invisible Touch'	'86	Genesis	<35>
'True Colors'	'83	Cindi Lauper	<35>
'Two Hearts'	'88	Phil Collins	<35>
'Something Happened on the Way to Heaven'	'89	Phil Collins	<35>
'There'll Be Sad Songs'	'86	Billy Ocean	<35>
'The Colour of Love'	'88	Billy Ocean	<35>
'Summer of '69'	'84	Bryan Adams	<35>
'It's Only Love'	'84	Bryan Adams	<35>
'I'll Be Loving You'	'88	New Kids	<35>
'Into the Night'	'80	Benny Mardones	<35>
'Midnight Rocks'	'80	Al Stewart	<35>
'Who'll Be the Fool Tonight'	'80	Larson/Feiten Band	<35>
'Angel Say No'	'80	Tommy Tutone	<35>
'Livin' On a Prayer'	'86	Bon Jovi	<35>
'Dead or Alive'	'86	Bon Jovi	<35>
'Even the Losers'	'80	Tom Petty	<35>
'Here I Go Again'	'87	White Snake	<35>
'Don't Let It End'	'83	Styx	<35>
'Holiday'	'83	Madonna	<35>
'Like a Virgin'	'84	Madonna	<35>
'Crazy for You'	'86	Madonna	<35>
'When Doves Cry'	'84	Prince	<35>
'Still Haven't Found'	'87	U2	<35>
'Don't Stop Believin'	'81	Journey	<35>
'Sign Your Name'	'87	Terrance T. D'Arby	<35>
'Gimme All Your Lovin'	'83	ZZTop	<35>
'Rough Boys'	'85	ZZTop	<35>
'Higher Love'	'86	Steve Winwood	<35>
'Legs'	'83	ZZTop	<35>
'Shot Through the Heart'	'84	Bon Jovi	<35>
'State of Grace'	'89	Billy Joel	<35>
'We Built This City'	'85	Starship	<35>
'Word Up'	'84	Cameo	<35>
'Where Are You Now'	'86	Jimmy Harnen	<35>
'Superstitious'	'88	Europe	<35>

Table 1. Continued

Song	Year	Artist	Pattern
'Back to Paradise'	'87	.38 Special	<35>
'Caught Up in You'	'82	.38 Special	<35>
'Fantasy Girl'	'80	.38 Special	<35>
'Trouble in Paradise'	'85	Huey Lewis	<35>
'Free Falling'	'89	Tom Petty	<35>
'Everybody Wants to Rule the World'	'85	Tears for Fears	<35>
'Back in the High Life'	'86	Steve Winwood	<35>
'Urgent'	'81	Foreigner	<35>
'Heat on the Street'	'89	Phil Collins	<35>
'All of My Life'	'89	Phil Collins	<35>
'High on You'	'86	Survivor	<35>
'Did It In a Minute'	'82	Hall and Oates	<35>
'Say It Isn't So'	'83	Hall and Oates	<35>
'Is This Love'	'84	Survivor	<35>
'One Heartbeat'	'87	Smokey Robinson	<35>
'Fast Car'	'88	Tracy Chapman	<332>
'Synchronicity II'	'83	The Police	<332>
'Mandolin Rain'	'86	Bruce Hornsby	<332>
'Don't Tell Me You Love Me'	'83	Night Ranger	<332>
'You Belong to the City'	'83	Glenn Frey	<332>
'I Go to Extremes'	'89	Billy Joel	<332>
'Vacation'	'82	The Go-Go's	<332>
'Power of Love'	'85	Huey Lewis	<332>
'She Goes Down'	'89	Mötley Crüe	<332>
'Don't Go Away Mad'	'89	Mötley Crüe	<332>
'Groovy Kind of Love'	'88	Phil Collins	<332>
'Here's to You'	'88	Billy Ocean	<332>
'Roll Me Away'	'82	Bob Seeger	<332>
'I Won't Hold You Back'	'82	Toto	<332>
'Fire in the Morning'	'80	Melissa Manchester	<332>
'Only a Lonely Heart Sees'	'80	Felix Cavaliere	<332>
'Even the Losers'	'80	Tom Petty	<332>
'Harvest Moon 269'	'85	U2	<332>
'Only the Young'	'83	Journey	<332>
'Who Can It Be Now'	'82	Men at Work	<332>
'Private Eyes'	'81	Hall and Oates	<332>
'Don't Ask Me Why'	'80	Billy Joel	<332>
'Suddenly Last Summer'	'83	The Motels	<332>
'Missing You'	'83	John Waite	<332>
'Jack and Diane'	'83	John C. Mellancamp	<332>
'Like No Other Night'	'86	.38 Special	<332>
'Lady of the Valley'	'87	White Lion	<332>
'Back Where You Belong'	'83	.38 Special	<332>
'Honestly'	'86	Stryper	<332>
'Cupid'	'80	The Spinners	<332>

Table 1. Continued

Song	Year	Artist	Pattern
'Surrender'	'82	Cheap Trick	<323>
'One Heartbeat'	'87	Smokey Robinson	<323>
'Easy Lover'	'84	Bailey/Collins	<323>
'Heaven'	'84	Bryan Adams	<323>
'Sign Your Name'	'87	Terrance T. D'Arby	<233>
'Caught Up in You'	'82	.38 Special	<233>
'Run to You'	'84	Bryan Adams	<233>
'True'	'83	Spandau Ballet	<233>
'Shoot to Thrill'	'80	AC/DC	<333322>
'Givin' the Dog a Bone'	'80	AC/DC	<333322>
'I've Done Everything for You'	'81	Rick Springfield	<333322>
'Did It In A Minute'	'82	Hall and Oates	<333322>
'Big Shot'	'80	Billy Joel	<333322>
'Dream Police'	'80	Cheap Trick	<333322>
'My Sharona'	'83	The Knack	<333322>
'Summer of '69'	'84	Bryan Adams	<333322>
'Photograph'	'83	Def Leppard	<333322>
'Runaway'	'84	Bon Jovi	<333322>
'For Those About to Rock'	'81	AC/DC	<333322>
'Your Love'	'85	The Outfield	<333322>
'Hard Habit to Break'	'84	Chicago	<333322>
'The Future's So Bright'	'83	Timbuk 3	<333322>
'Melt with You'	'83	Modern English	<333322>
'Karma Chameleon'	'83	Culture Club	<333322>
'Lady of the Valley'	'87	White Lion	<333322>
'Crazy for You'	'86	Madonna	<333322>
'Desire'	'88	U2	<333322>
'Tell Me'	'87	White Lion	<333322>
'Back to Paradise'	'87	.38 Special	<333322>
'Teacher, Teacher'	'84	.38 Special	<333322>
'Angel Say No'	'80	Tommy Tutone	<333322>
'Panama'	'83	Van Halen	<333322>
'Runnin' Down a Dream'	'89	Tom Petty	<333322>
'Simply Irresistible'	'88	Robert Palmer	<333322>
'Dirty Mind'	'80	Prince	<333322>
'Private Eyes'	'81	Hall and Oates	<333322>
'Don't Let It End'	'83	Styx	<333322>
'Rosanna'	'82	Toto	<333322>
'Peace in Our Time'	'89	Eddie Money	<333322>
'Don't Ask Me Why'	'80	Billy Joel	<333322>
'Faithfully'	'83	Journey	<233332>
'Simply Irresistible'	'88	Robert Palmer	<233332>
'Smuggler's Blues'	'85	Glenn Frey	<233332>
'If I'd Been the One'	'83	.38 Special	<233332>
'While You See a Chance'	'80	Steve Winwood	<233332>

Table 1. Continued

Song	Year	Artist	Pattern
'Crazy Train'	'81	Ozzy Osbourne	<233332>
'Listen to Her Heart'	'80	Tom Petty	<233332>
'Ragdoll'	'85	Aerosmith	<233332>
'Lay Your Hands On Me'	'88	Bon Jovi	<233332>
'Back In Black'	'80	AC/DC	<233332>
'Love Walks In'	'86	Van Halen	<233332>
'Vacation'	'82	The Go-Gos	<233332>
'Looks That Kill'	'83	Mötley Crüe	<233332>
'Message In A Bottle'	'80	The Police	<3445>
'Run to You'	'84	Bryan Adams	<3445>
'State of Grace'	'89	Billy Joel	<3445>
'I Wanna Go Back'	'86	Eddie Money	<34522>
'I Don't Wanna Lose You'	'83	REO Speedwagon	<344(5)>
'Power of Love'	'85	Huey Lewis	<344(5)>
'That Ain't Love'	'86	REO Speedwagon	<344(5)>
'Get Away'	'82	Chicago	<344(5)>
'Fortress Around Your Heart'	'85	Sting	<3445>
'Madelaine'	'88	Winger	<3445>
'Wrapped Around Your Finger'	'83	The Police	<3445>
'Valerie'	'82	Steve Winwood	<3445>

title, the approximate year of its release, the group or act that performed the song, and the accent pattern used in the song. In some cases, the same song appears on this list twice, meaning it features two different accent patterns. (This is not surprising, as it is often the case that short, introductory patterns provide a glimpse of a longer pattern to be heard later in the song. This is certainly true in Bryan Adams' hit, 'Summer of '69', which uses a clear <35> pattern during the guitar introduction to foreshadow the powerful <333322> pattern that punctuates the instrumental climax. Billy Joel's 'Don't Ask Me Why' provides a similar example, except the <332> pattern stated in the introductory keyboard riff turns immediately into the <333322> that sets the text of the verses.) Clearly, these patterns were 'stock' gestures that 1980s musicians used over and over again. This alone, however, does not necessarily make them hooks. In fact, one could argue that due to their pervasiveness, it becomes especially difficult to utilise them as a hook for a song. As with most musical definitions, context is the key, a topic to which we now turn.

In general, there are three basic ways that the accent patterns form hooks. The songs listed in Table 2 demonstrate the first way. In these songs, the delivery of the title clearly serves as the hook, with one of the accent patterns providing the rhythmic portion of this equation. In songs such as Van Halen's 'I'll Wait', and Phil Collins' 'Two Hearts', the brevity of the two-syllable title lends itself well to the two accents of the <35> pattern. Indeed, in both of these song titles, this combination forms a simple but effective hook. In other cases, the text/accent relationship is somewhat more complex. If a <35> pattern is in force, but the song's title contains more than two syllables, the accents typically fall on the first and last syllables, quickly working in the

Table 2. Songs with clear title hooks.

<35>
Van Halen, 'I'll Wait'
Phil Collins, 'Two Hearts'
Benny Mardones, 'Into the Night'
Styx, 'Don't Let It End'
Billy Joel, 'State of Grace'
Journey, 'Don't Stop Believin'
Bon Jovi, 'Shot Through the Heart'
Terrance T. D'Arby, 'Sign Your Name'
.38 Special, 'Back to Paradise'
Cheap Trick, 'Dream Police'
Def Lepard, 'Photograph'
Van Halen, 'Panama'
Hall and Oates, 'Private Eyes'
<332>
Billy Joel, 'I Go To Extremes'
Toto, 'I Won't Hold You Back'
Journey, 'Only the Young'
Men at Work, 'Who Can It Be Now?'
<323>
Cyndi Lauper, 'True Colors'
Smokey Robinson, 'One Heartbeat'
Collins/Bailey, 'Easy Lover'
<333322>
Rick Springfield, 'I've Done Everything for You'
Hall and Oates, 'Did It In a Minute'
Billy Joel, 'Big Shot'
<233332>
Bon Jovi, 'Lay Your Hands On Me'
AC/DC, 'Back in Black'
Van Halen, 'Love Walks In'
Mötley Crüe, 'Looks That Kill'

others between the accents on one and the 'and' of two: 'In-to the **Night**', 'Don't Let It **End**', 'Shot Through the **Heart**'. Journey's 'Don't Stop Be-leiv-in' uses two bars of <35> pattern to set the title.

Slightly longer title hooks set entire phrases of text, not just the actual title. Billy Joel's song, 'I Go To Extremes', provides one such example involving two successive <332> patterns: '(Darlin' I) **Don't Know Why I Go To Ex-(tremes)**'. In similar fashion, Toto's 'I Won't Hold You Back' also sets the entire title with two <332> patterns: '(You) **Know I Won't Hold You Back (Now)**'. In each of these songs, the verses build quite nicely to the chorus without employing the accent pattern used to set the title hook. Thus, the impact is that much more effective. (Incidentally, Toto's 'I Won't Hold You Back' provides another nice example of shorter patterns building to longer ones, as the <332> used in the title lyric builds to a <333322> to cap the guitar solo later in the song.)

While several of the above songs use two statements of a one-measure pattern to create a two-measure title hook, there are also examples that feature strong title hooks using a single two-measure accent pattern, such as the <333322>. Rick Springfield's hit, '**I've Done E-v'ry-thing For (You)**', is one such instance. Aside from the final 'You', each syllable of the title is set with an accent in the pattern. This passage is made even stronger by having all instruments play in unison with the vocals, thus forming a powerful hook. Billy Joel's 'Big Shot' follows a similar layout, using a <333322> pattern in a syllabic setting of the title lyric: '(You had to be a) **big shot, didn't ya, ya had to** (open up your mouth)'. On the other hand, 'Did It in a Minute', by Hall and Oates, uses the four groups of three to highlight a particular portion of the title lyric: '**Did it, ya did it, ya did it, ya did it, ya did it in a** (minute)'.

Several songs use the <(2)33332> pattern in a similar fashion. Again, it is most common for the setting to be syllabic, with one word of the text set against one accent of the pattern: '**She's got looks that kill**' (Mötley Crüe), '**Lay your hands on me**' (Bon Jovi), '**Love comes walk-in' in**' (Van Halen). AC/DC's 'Back in Black' is an exception. Strong emphasis is given to the word 'Back', repeating the first syllable, seemingly mimicking the same <(2)33332> pattern heard earlier in the guitar: '**Bay-hay-hay-hay-ack**'. Each of these title lyrics uses the metric dissonance of this pattern to add to the power of their hard-rock style.

The above passages are quite clearly hooks, and in most cases it is not difficult to align this reading with Burns' definition. As was stated earlier, Burns argues that change, or 'modulation', is the defining element of a hook. Keil suggests similar tenets with his distinction between 'significant gestures' and the 'vital drive' which they interrupt. The songs in Table 2 achieve this modulation in different ways. In cases such as 'I've Done Everything For You', and 'I Go to Extremes', the chorus (and song's title) is set off clearly from the rest of the piece – in part due to the accent pattern – and forms a strong hook. In other cases the change is more subtle, created simply by the fact that the text delivers the song's title. Indeed, songs like 'Sign Your Name' and 'Into the Night' do not have dramatic shifts in texture that mark the start of the chorus, but the delivery of the song's title creates a hook nonetheless.

There are several cases that illustrate a second use of accent patterns to create potential hooks. In this group of songs, the hook is not unequivocally one particular passage. The title still is set strongly and does serve as a hook, but in each case there is also another prominent gesture using one of the patterns, that is certainly a 'significant gesture', and could be considered a hook as well. Robert Palmer's 1988 hit, 'Simply Irresistible', provides an excellent example. Certainly the strong setting of the song's title during the chorus creates a memorable hook for this song. But according to Palmer, it is actually the passage immediately prior to title lyrics that creates the hook. According to Palmer:

I had ['Simply Irresistible'] lying around for three years and I wanted to finish it off but could never find the release line, the hook, until I hit on the line 'But now I find her . . .', with the pause and the resolution. A little thing like that makes the difference between an idea and the complete song. (Palmer 1989, p. 16)

The 'pause' and 'resolution' to which Palmer refers is where the <333322> pattern initiated by the drum solo occurs, immediately after the line, 'But now I find her . . .'. Thus, it seems plausible that two hooks are at work here. The song's title is memorable with or without the drum passage, but the drum passage itself has quite distinct characteristics as well.

Several other songs from the 1980s use the <333322> pattern to create a significant gesture just prior to the chorus. In Cheap Trick's 'Dream Police', for instance, the three-against-two feel heard during the pre-chorus jarringly interrupts the underlying groove. The text, '**E-v'ry sin-gle night they're . . .**', is set in a <333322> pattern, with all instruments and the vocals in unison, thus effectively expressing the panic of being driven insane by the relentlessness of 'those men inside my brain'. This gesture certainly has the significance of a hook. At the same time however, the song's chorus, 'The Dream Police', is memorable as well, suggesting multiple hooks.

The Knack's 'My Sharona' provides another example. Here, the pre-chorus features a rising bass line that builds to a climax, where the six syllables of the line, '**My-My-My-My-ah-woo!**', are set with the metric dissonance of a <333322> pattern. The four utterances of 'My', spread out over a two-measure span during this gesture, create a nice elongation of the much shorter title hook, 'M-M-M-My Sharona', heard later in the chorus. In another example, Hall and Oates' hit, 'Did It in a Minute', employs two cycles of the <333322> paradigm during a build up to the chorus. The accent pattern combines with leaping tritones and descending sixths as the harmony builds to the dominant in preparation for the song's chorus (which is itself set with a <333322> pattern).

Just as these examples use a pre-chorus hook to build to the chorus, so do other songs use similar gestures for other formal purposes. Chicago's 'Hard Habit to Break', for example, uses a <333322> pattern just prior to a short instrumental interlude that leads back to the refrain: '**I just can't go ah-on**'. The same accent pattern occurs midway through The Who's hit, 'Who Are You', again providing material just prior to an instrumental interlude: '**Who wah, who wah, who wah**'. The soft, somewhat vacuous repetition of a truncated version of the song's title makes a nice segue into the keyboard solo. In 'Peace in Our Time', from Eddie Money, and Aerosmith's hit, 'Ragdoll', the pattern occurs within the verse: '(Never gonna) **tear down the walls and turn the** (prison into stone)'; '**Yes I'm mo-vin**'. A final example comes from 'Summer of '69', by Bryan Adams, where the ending of the bridge explodes into a very expressive and powerful <333322> pattern in the guitar and keyboards, again leading back to the verse.

It is clear, then, that many of these songs employ accent patterns for contrast or to highlight certain portions of the song. And in certain cases, particularly 'Dream Police' and 'My Sharona', they reinforce Burns' notion that contrasting sections – 'significant gestures', as Keil would call them – make strong hooks. The fact that many of these gestures use the same or similar syncopated accent patterns makes the observation that much more interesting. Significantly, each of these contrasting passages features two-measure accent patterns, simply because these allow for a bolder, more expressive musical gesture than do the one-measure patterns. Indeed, the momentary metrical dissonance built into these longer patterns seems tailor-made for singular, high-impact, and memorable gestures.

But as it turns out, these same accent patterns often function in a different fashion, one that seems to conflict with Burns' ideas about repetition. It is important to remember here that Burns dismisses repeated rhythmic patterns as hooks. To quote him:

Rhythm almost always involves repetition – so to look for rhythmic hooks that operate by repetition would be unproductive. If a straight 3/4 rhythm repeats throughout an entire song, is it meaningful to speak of the rhythm as a hook? (Burns 1987, p. 6)

In general, Burns' point is well taken; change is an important element of any music, and the gestures that create those changes are often what we remember most about a particular piece.

That said, repetition perhaps should not be dismissed so quickly as a device for hooking listeners' attention. Indeed, repetition in various forms can be noteworthy in certain contexts as well. What if, for instance, the straight 3/4 rhythm Burns mentions is repeated within a song in 4/4, as in Led Zeppelin's 'Kashmir'? Certainly the conflict between John Bonham's heavy duple foundation in the drums and the rising triple pattern in the guitar is part of the hook of this tune, despite its repetition.

A third means of creating hooks using these patterns is to use them as the basis for the main groove. The <333322> pattern, for example, with its extended triple feel, is often repeated extensively as the main groove, or 'vital drive' of a particular song. But despite its repetition, its unique structure commands attention and hooks the listener. Consider songs such as 'Your Love', by The Outfield, where the guitar provides a distinct <333322> riff from the song's outset. Since the title lyric is not set as a strong hook in this song, the guitar pattern takes on that much more prominence. Billy Joel's 'Don't Ask Me Why' also features a clear <333322> pattern throughout most of the song's verses: '**All the wait-ers in your** (grande café)'. Both of these songs use the <333322> pattern in a repetitious manner, but its hook is strong nonetheless. Further examples using the <3445> come from Sting's 'Fortress Around Your Heart' (in the accompaniment throughout, but most prominently in the chorus), and the guitar riff in 'Run to You', from Bryan Adams.

Repetitious rhythmic patterns work best as hooks when combined with a repeated pitch pattern. This is a point that Burns picks up on, as he states that 'the establishment of a rhythm may be combined with the establishment of a riff, that is, a repetitive melodic hook' (Burns 1987, p. 6). This is especially true when we consider how one-measure rhythmic patterns become part of the hook when combined with pitch patterns of two or more measures. Example 4 shows several such cases in which short rhythmic patterns combine with two-bar pitch patterns. Michael Jackson's pop hit, 'Billie Jean', is a clear example using a <35> accent pattern. The resulting two-bar pattern may not be as strong a hook as his delivery of the song's title, but the syncopated pattern is a memorable one. This type of pitch/rhythm combination is common for <332> patterns as well (see Examples 4(b)–(e)). Tracy Chapman's 'Fast Car' and John Cougar Mellancamp's 'Jack and Diane' each feature a <332> rhythmic pattern with a two-bar pitch sequence. Neither of these songs uses the title lyric as a hook, so the patterns cited here make viable options. The final example here is from The Go-Gos' pop hit, 'Vacation', which features a nice instrumental interlude that recalls the syncopations of song's introduction and provides contrast. The pitch content here is noteworthy as well, featuring a four-bar pitch pattern that completes stepwise motion that surrounds G# (Example 4(d)).

This example from The Go-Gos points to an important point about these pitch/rhythm combinations. While this study is focusing on a few basic rhythmic patterns used to form hooks, the 'double-neighbour' pattern from 'Vacation' is one example of a recurring pitch pattern as well. Quite often, the <333322> and <(2)33332> patterns will set the four groups of three using the double-neighbour pattern. Bon Jovi's 'Runaway' is a clear example (see Example 5(a)). During the keyboard riff at the opening of the song, the outer-most parts remain unmoving, but the inner part presents double-neighbour motion around the chordal third, C. In AC/DC's 'For Those About to Rock', a very long, very distorted guitar solo repeats the pattern

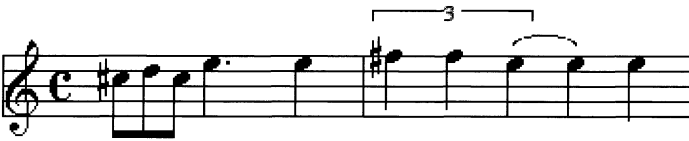
(a) Michael Jackson, 'Billie Jean'



(b) Tracy Chapman, 'Fast Car' (guitar)



(c) John Cougar Mellancamp, 'Jack and Diane' (guitar)



(d) The Go-Go's, 'Vacation' (keyboard)

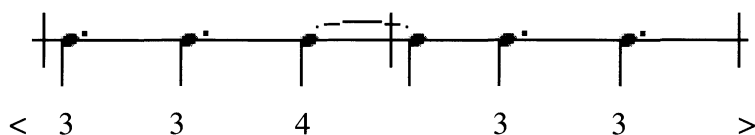


Example 4. Pitch/rhythm combinations.

shown in Example 5(b) several times before the vocals enter. As mentioned earlier, 'Summer of '69' also sets a <333322> pattern in this manner.

Examples 5(c)–(e) show that the four groups of three in the <(2)33332> pattern are also often set as some type of double-neighbour motion around a particular tone, usually the choral third of the tonic. This is the case in Journey's rock ballad 'Faithfully', where the pattern shown in Example 5(c) is heard throughout the song, both in the instruments, the vocals, and even in the guitar solo. 'If I'd Been the One' (Example 5(d)) from .38 Special is quite similar, except the pattern is heard only during the introduction and verses. The rhythm guitar heard during the verses of 'Crazy Train' (Example 5(e)) and 'Ragdoll', mentioned earlier, provide further examples. Indeed, the unity of a distinctive accent pattern and a clear pitch pattern creates a strong hook in these songs, despite the repetition of the gesture.

The above examples point to several insights about the nature of rhythmic hooks in 1980s rock music. While it is important to note that syncopation is a prominent part of rock rhythm, it is equally important to realise that songwriters used syncopation to form accent patterns, patterns which are found again and again in a multitude of songs. Furthermore, the fact that these patterns function equally well as contrasting, 'significant gestures', and as part of the main groove, forces us to think carefully about



Example 7. REO Speedwagon, 'Don't Let Him Go'.



Example 8. The Tubes, 'She's a Beauty'.

Unique rhythmic hooks from the 1980s

The accent patterns discussed thus far are those that have a wide appeal and are found in numerous songs. Several of their general principles, however, help to create strong hooks in patterns unique to particular songs. For instance, the main keyboard riff from Journey's ballad, 'Who's Crying Now', features a two-measure pattern with an unaccented two-group on the downbeat, placing the initial accent on beat 2 (just as we saw in the <(2)33> and <(2)33332> patterns presented above). But instead of synco-pating through beat three of the initial measure, this accent on beat 2 acts as a pickup to another accent *on* beat 3, where a three-group initiates an anticipation of the next downbeat, which is followed by an anticipation of the next strong beat 3 (see Example 6). The result is an interesting <(2)2 3 4 5> grouping, suggesting an additive element, as each accented group grows by one eighth-note. The pattern permeates all parts of the song and is often reinforced by the text, particularly during the chorus' setting of 'won-der who's cry-in' now'.

Other songs demonstrate further combinations involving the four groups of three seen in several examples above. Example 7 shows the rhythmic pattern comprising the main groove of REO Speedwagon's 'Don't Let Him Go'. This pattern features a four group sandwiched between three groups, suggesting a <332233> pattern. This is interesting because it implies yet another rotated version of the prototype <333322>, shown earlier in Example 3. 'She's a Beauty', from The Tubes, uses this pattern as well in the first measure of a catchy two-bar pattern, the rhythm of which is shown in Example 8. The introduction to 'Reminiscing' by Little River Band features a similar rhythmic pattern, but uses a much more complex four-bar pitch pattern (see Example 9). The first four-bar phrase of the melodic line presents a <33433> eighth-note grouping structure, followed by a <332>. This syncopated melodic pattern is quite prominent against the steady bass, which lays down a foundation of half notes on one and three throughout the passage. The rhythmic pattern of the first phrase is repeated in the next phrase before giving way to a pair of quarter-note <35> groupings in the final phrase. The harmonic content is equally complex, beginning in E $\flat$ and moving through a cycle of fifths before landing on the tonic D.

Two final examples are noteworthy for their length. Shunning the tendency for duple divisions at all levels of metric structure, Kansas incorporates a three-measure pattern into their hit, 'Carry on Wayward Son' (see Example 10). The passage shown here acts as part of a long instrumental introduction. It begins with the familiar

Example 9. Little River Band, 'Reminiscing' (intro).

Example 10. Kansas, 'Carry on Wayward Son'.

Example 11. Van Halen, 'Jump' (keyboard riff).

two-bar <333322> pattern, but tacks on an extra measure with heavy accents on the 'and' of beats one and three, creating a three-bar pitch and rhythmic group, a rarity in this genre. Finally, as Example 11 shows, the opening keyboard solo from Van Halen's 'Jump' presents a recurring pitch and rhythm pattern four full measures in length. As

Table 3. Songs prior to 1980.

Song	Artist	Pattern
'I'm Happy Just to Dance with You'	The Beatles	< 3 5 >
'Chain Lightning'	Steely Dan	< 3 5 >
'Breakdown'	Tom Petty	< 3 5 >
'Don't Do Me Like That'	Tom Petty	< 3 5 >
'Even the Losers'	Tom Petty	< 3 5 >
'Babe	Styx	< 3 5 >
'Live and Let Die'	Paul McCartney	< 3 5 >
'The End'	The Beatles	< 3 5 >
'Let's Get It On'	Marvin Gaye	< 3 5 >
'Listen to the Music'	The Doobie Brothers	<332>
'Swingtown'	Steve Miller Band	<332>
'True Fine Love'	Steve Miller Band	<323>
'Ballad of John and Yoko'	The Beatles	<323>
'Let's Get It On'	Marvin Gaye	<323>
'Come Sail Away'	Styx	<323>
'Dance the Night Away'	Van Halen	<323>
'I'll Cry Instead'	The Beatles	<233>
'Doctor Robert'	The Beatles	<233>
'Things We Said Today'	The Beatles	<233>
'The Word'	The Beatles	<233>
'Tomorrow Never Knows'	The Beatles	<233>
'Roll Me Away'	Bob Seeger	<332>
'You'll Accomp'ny Me'	Bob Seeger	<35>
'Hollywood Nights'	Bob Seeger	<333322>
'Still the Same'	Bob Seeger	<35>
'The Twist'	Chubby Checker	<35>
'Hound Dog'	Elvis Presley	<332>
'Double Vision'	Foreigner	<35>
'Head Games'	Foreigner	<35>
'Urgent'	Foreigner	<35>
'Dirty White Boy'	Foreigner	<35>
'Peace of Mind'	Boston	<332>
'Don't Look Back'	Boston	<35>
'Chameleoan'	Steely Dan	<23333>
'Roll with the Changes'	REO Speedwagon	<233332>
'Good Lovin'	Young Rascals	<333322>
'Who Are You'	The Who	<333322>
'Hotel California'	Eagles	<333322>
'Here Comes the Sun'	The Beatles	<333322>
'Sunshine of My Life'	Stevie Wonder	<333322>
'Satisfaction'	Rolling Stones	<233332>
'Live and Let Die'	Paul McCartney	<233332>
'Honky Cat'	Elton John	<233332>
'Never Be the Same'	Christopher Cross	<3445>
'Sailing'	Christopher Cross	<3445>
'Layla'	Clapton	<3445>

is typical of several other patterns, it begins with an unaccented two-group, then moves through a relatively complex mix of two- and three-groups before a five-group marks the end of the pattern and prepares the listener for its repetition. The sense of syncopation is quite strong due to the anticipations of all strong beats after beat one of measure two. In fact, after the second measure, the only beat with an accent is beat two of measure three, which sounds like an internal goal of the passage, particularly due to its climactic property of presenting the highest pitch.

The accent patterns in other eras of rock

As mentioned earlier, it should not come as a surprise that multitudes of songs from the 1980s would share a few basic accent patterns. The corporate influence on the music industry became overwhelming during this era; the need to sell records led to the homogeneity that Burns alluded to in his study on hooks. It is important to note, however, that just because the accent patterns were prominent during this time does not mean that they were new phenomena. In fact, each pattern cited in Examples 2 and 3 has precedence in several songs released prior to 1980.⁸

To illustrate this, Table 3 lists numerous songs released between 1956 and 1979. Several interesting points emerge from this table. For example, there are several more instances of the <323> and <(2)33> one-measure patterns than shown in Table 1. This is due in large part to The Beatles affinity for the <(2)33> pattern in their guitar riffs. Indeed, the rhythmic quality of the riffs from 'Doctor Robert', 'Things We Said Today', and 'The Word', is almost interchangeable, featuring the unaccented down-beat, but the strong emphasis on beat 2 and the 'and' of 3. Is it possible that these patterns 'fell by the wayside' in the homogenous land of 1980s music? If so, why? And when we do hear them in an 1980s tune, is it meant as a throw-back to earlier eras of rock? Clearly there is not enough evidence here to support these claims entirely, but the questions may be worth exploring.

A similar avenue for further investigation involves the relationship of the accent patterns to the notion of hook. As shown earlier, several songs from the 1980s create a strong hook by setting the title lyric with one of the accent patterns. But was this tendency as strong prior to the 1980s? When we look for similar instances in songs from Table 3, only a few examples emerge. Certainly 'Babe', the rock ballad from Styx, makes a clear hook by setting '(You Know It's) **You, Babe**', with a clear <35> pattern. Bob Seeger's <35> setting of '**Still the Same**' provides another example. Foreigner's '**Dir-ty White Boy**', is a rare <323> hook. And The Rolling Stones <(2)33332> setting of '**I Can't Get No-oh, Sa-tis-fac-tion**', is a clear hook as well. Beyond that, however, one is hard-pressed to find instances where the pattern could be considered a hook. Many of the examples featuring the <333322> pattern employ them simply for contrast, as in the Young Rascals' 'Good Lovin', the guitar solo at the end of 'Hotel California', and the keyboard passage during the introduction of Stevie Wonder's 'Sunshine of My Life'. These preliminary findings suggest further that the 1980s were a time when the patterns identified above were at their most pervasive, and when their integration into musical hooks was most prominent.

Endnotes

1. While I do think Keil's points apply to rock as well as jazz, there are some interesting and important differences. For one, as Keil states, the pulse in jazz is often subjective or implicit. This

is not the case in rock, where it is quite explicit. Also, while jazz musicians tend to lay behind the beat, we will see that the patterns discussed herein almost all feature *anticipation* of strong beats.

2. For views of repetition and its role in popular music see, among others, Middleton (1983, pp. 235–70) and Monson (1999, pp. 31–65).
3. David Temperley used this system quite effectively in his study on the perception of syncopation (Temperley 1999, pp. 19–40).
4. While I believe that these represent the most common accent patterns found in rock songs, they are by no means the *only* patterns to be found.
5. As mentioned earlier, it is interesting to note that these displacements are almost exclusively *anticipating* stronger beats, not holding them over as in a suspension. That's not to say that 'forward' displacement doesn't happen in rock. There are certainly suspensions, and some repertoires stress the off-beat in a way that might be considered dragging, as in a Texas shuffle, say Stevie Ray Vaughan's 'Pride and Joy'. In rock, however, the tendency is to anticipate, thus driving the motion forward.
6. In his article on Caribbean music, Floyd identifies this exact same rhythm as the *tresillo* rhythm and addressed its connection to and origins in the African time-line patterns of sub-Saharan Africa (Floyd, Jr 1999, pp. 1–38).
7. The notion of rotation has also been applied to rock harmony by Guy Capuzzo (2001). Capuzzo suggests that some common rock progressions can be thought of as rotations of familiar circle of fifths patterns. For example, the fifth pattern A–E–B–F# can be rotated once to create F#–A–E–B, resulting in a I–bIII–bVII–IV progression, heard in such songs as 'Run to You', by Bryan Adams, and 'Fly Away', by Lenny Kravitz. While it seems more intuitive to conceptualise these patterns in terms of their displacement of underlying beats, the rotation metaphor is also valid.
8. The presence of these patterns in other repertoires is a topic for another study. In addition to my earlier note about their origins in African time-lines and incorporation into music of the Caribbean, I have found examples from jazz, ragtime, Hawaiian music, Klezmer music, and Latin American tango.

References

- Burns, G. 1987. 'A typology of "hooks" in popular records', *Popular Music*, 6/1, pp. 1–20
- Capuzzo, G. 2001. 'Rotation as a model for rock chord progressions', paper presented at the Florida State Theory Forum, Tallahassee
- Floyd, Jr., S.A. 1999. 'Black music in the Circum-Caribbean', *American Music*, 17/1, pp. 1–38
- Keil, C. 1994. 'Motion and feeling through music', in *Music Grooves*, by Charles Keil and Steven Feld (Chicago)
- Lerdahl, F., and Jackendoff, R. 1983. *A Generative Theory of Tonal Music* (Cambridge, MA)
- Middleton, R. 1983. ' "Play it again Sam": some notes on the productivity of repetition in popular music', *Popular Music*, 3, pp. 235–70
1990. *Studying Popular Music* (Open University Press)
- Monson, I. 1999. 'Riffs, repetition, and theories of globalization', *Ethnomusicology*, 43/1, pp. 31–65
- Murphy, S. 2002. '(Re)solving one kind of metric dissonance', paper presented at the National Meeting of the Society for Music Theory, Columbus
- Palmer, R. 1989. Quoted from the liner notes to *Robert Palmer: 'addictions' volume 1*, Island Records
- Temperley, D. 1999. 'Syncopation in rock: a perceptual perspective', *Popular Music*, 18/1, pp. 19–40